

Dragravity: Wearable, Reconfigurable, Dynamic Weight-Shifting Waterbags to Enhance Full-Body Weight Sensations

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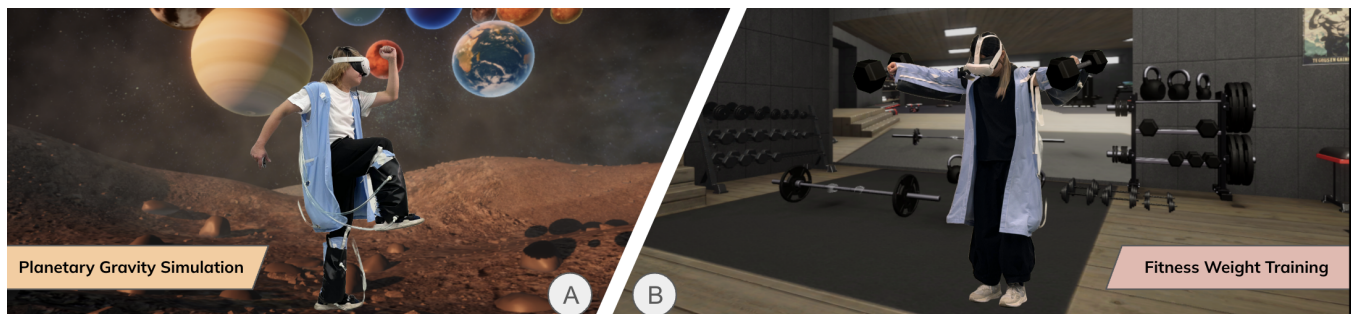


Figure 1: We present Dragravity, a wearable device capable of dynamically shifting weight. (A) Experience the gravity feel on different planets. (B) Operating weight training in a virtual gym, simulating different weight levels with water.

Abstract

Many studies have explored weight feedback in virtual reality (VR), but few can provide sufficient full-body weight sensations. We present Dragravity, a dynamic weight-shifting haptic proxy using wearable, reconfigurable waterbags. Our waterbags can be worn on arms, lower body, or legs using elastic bands to distribute weight across different body parts. This enhances the user's full-body weight sensation, enabling application possibilities with the haptic proxy. The waterbags can simulate weight sensation in VR using water with a maximum additional weight of 10.4 kg. To demonstrate, we designed two applications. The first is planetary exploration, where users can experience gravitational sensations on planets with the dynamic full-body weight sensation. The second is fitness weight training. By leveraging dynamic weight, users can experience various types of weight training on different muscles.

CCS Concepts

• **Human-centered computing** → **Interaction paradigms**; Virtual Reality.

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Keywords

Liquid - based Haptic, piping System, Weight Sensations, Gravitational Sensations, Virtual Reality

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1 Introduction

To enhance immersion in VR, many studies have attempted to incorporate haptic feedback by simulating the sensation of weight. [Kalus et al. 2024] uses a water pump to transfer water to a handheld controller, with a maximum weight of 1 kg, and allows the water pump to be detached from the device for flexible usage. [Wang et al. 2022] and [Cheng et al. 2018] simulate weight changes by adjusting the amount of liquid in the device, with a maximum weight of 0.9 and 1.5 kg. [Niiyama et al. 2014] use liquid metal to simulate weight changes, providing various sensations of densities. [Chen et al. 2022] showcases a wearable device that increases weight using water, with a maximum weight of 1.8 kg. Apart from research that directly increases weight, [Yamashita et al. 2018] simulates the feeling of elongated forearms by shifting the position of weight in the wearable device. [Nith et al. 2023] creates an accelerated and decelerated jumping sensation by moving a wearable weight up and down,

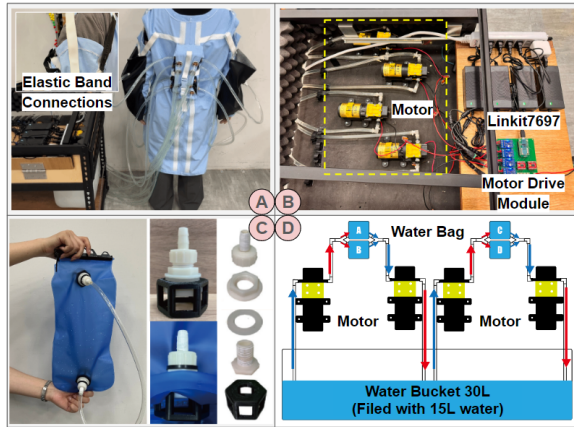


Figure 2: (A) Device Structure and Elastic Bands Connections, (B) Microcontroller, Motor Driver Modules, and Motors, (C) Water Bag Structure, (D) Water Pipeline System

altering the user's perception of jumping. Among these studies, we found that there are limitations in maximum weight capacity, which could be the reason why interactive devices rarely simulate weight on the legs since the lower body is normally stronger and requires more weight to create sufficient haptic sensation. Therefore, we present Dragravity (Fig. 1), wearable, reconfigurable waterbags for dynamic weight-shifting haptic proxy. Our waterbags can be worn on arms, lower body, or legs with elastic bands to distribute the weight across different body parts, enhancing the user's full-body weight sensation to enable more application possibilities with the haptic proxy, which can simulate the weight sensation in VR using water with a maximum additional weight of 10.4 kg.

2 DESIGN AND IMPLEMENTATION

To achieve the haptic proxy with full-body weight perception, we created a wearable device capable of reconfigurable, dynamic weight-shifting waterbags (Fig. 2A). This device consists of four 3-liter water bags connected with water pipes (Fig. 2C), wearable on the arms, lower body, or legs. A Linkit7697 sends signals to four 17A high-power motor driver modules, which drive four high-power motors (12V/5A, 8.0L/min) (Fig. 2B) to transfer water from a reservoir to the corresponding water bags to increase weight or extract back to the reservoir to decrease weight (Fig. 2D) to provide full-body weight sensations with adjustable weight. Dragravity features a detachable structure design that uses elastic bands, allowing arm-length adjustment and enabling the sleeve parts to be detached and worn on the legs. Through the stretchability of the elastic bands, the heavier the water bags on the arms, the longer the distance between the bags and the shoulders due to gravity, increasing the lever arm and thus the perceived pulling force, leading to an increased weight sensation. Conversely, in the lighter water bags, the elastic bands retract them closer to their original position, shortening the lever arm and reducing the perceived pulling force and gravity sensation. In performance tests, the water delivery rate from the reservoir to the gravity suit was 138.8 grams/second, while the extraction rate was 109.6 grams/second. The total weight gain can reach 10.4 kg, and the perceived gravity can be even greater through the lever effect.

3 Application

To demonstrate the concept of our device, we use Unity to create two VR scenarios. Wearing the VR head-mounted display and noise-canceling headphones, users can experience full-body haptic feedback from Dragravity. The first scenario is planetary exploration. Users can choose to visit different planets and explore various feels of gravity across Earth (9.8 m/s^2), Mars (3.7 m/s^2), or Jupiter (24.8 m/s^2). In this scenario, the device is worn on the lower body and legs, adjusting the volume of water to change the weight of the wearable device and simulate senses of gravity from different planets. The second scenario is fitness weight training. Users can engage in weight training exercises like kettlebell swings, dumbbell lateral raises, deadlifts, and battle ropes. Users wear the water bags on the corresponding body parts based on the specific exercise. Dragravity adjusts the volume of water in the bags according to the desired weight load the user selects to simulate realistic weight training in VR. By providing full-body haptic feedback, we aim to use these two scenarios to offer enhanced immersive experiences with a dynamic sense of weight and gravity.

4 DISCUSSION AND FUTURE WORK

Dragravity is a dynamic weight-shifting haptic proxy using wearable and reconfigurable waterbags that can be worn on different body parts. Users can perceive the dynamic full-body weight sensation from waterbags and the tension of elastic bands, experiencing the feel of gravity changes and thereby increasing immersion in VR. We designed two VR scenarios to introduce the applications of this wearable device. In the future, we will conduct user studies to investigate the effect of full-body haptic feedback created by the dynamic weight simulation in Dragravity. Additionally, we will explore the differences in gravity perception across different body parts to optimize the device structure for a better experience.

Acknowledgments

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